gravitons: Hypothetical messenger particles of the gravitational force.

grand unification: Theory attempting to unify the strong, weak, and electromagnetic forces.

Higgs field: See *electroweak Higgs field.*

Higgs field vacuum expectation value: Situation in which a Higgs field acquires a nonzero value in empty space; a Higgs ocean.

Higgs ocean: Shorthand, peculiar to this book, for a Higgs field vacuum expectation value.

Higgs particles: Finest quantum constituents of a Higgs field.

horizon problem: Challenge for cosmological theories to explain how regions of space, beyond each other's cosmological horizon, have nearly identical properties.

inertia: Property of an object that resists its being accelerated.

inflationary cosmology: Cosmological theory incorporating a brief but enormous burst of spatial expansion in the early universe.

inflaton field: The field whose energy and negative pressure drives inflationary expansion.

interference: Phenomenon in which overlapping waves create a distinctive pattern; in quantum mechanics, involves seemingly exclusive alternatives combining together.

Kaluza-Klein theory: Theory of universe involving more than three spatial dimensions.

Kelvin: Scale in which temperatures are quoted relative to absolute zero (the lowest possible temperature, −273° on the Celsius scale). luminiferous aether: See *aether.*

M-theory: Currently incomplete theory unifying all five versions of string theory; a fully quantum mechanical theory of all forces and all matter.

Mach's principle: Principle that all motion is relative and that the standard of rest is provided by average mass distribution in the universe.

Many Worlds interpretation: Interpretation of quantum mechanics in which all potentialities embodied by a probability wave are realized in separate universes.

messenger particle: Smallest "packet" or "bundle" of a force, which communicates the forces' influence.

microwave background radiation: See *cosmic microwave background radiation.*

negative curvature: Shape of space containing less than the critical density; saddle-shaped.

observable universe: Part of universe within our cosmic horizon; part of universe close enough so that light it emitted can have reached us by today; part of universe we can see.

open strings: Filaments of energy in string theory, in the shape of snippets. *p*-brane: Ingredient of string/M-theory with p-spatial dimensions. See also D-brane.